Smooth bounded nonlinear regression need not satisfy covariance domination in infinite dimension

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Status. Candidate full counterexample; likely valid subject to human review. It answers negatively the natural smoothness-only interpretation of the covariance-bound question of Mollenhauer–Mücke–Sullivan [1].

1 The question and the answer

For Hilbert-valued X, Y, the source uses the uncentred covariances $C_{YX} = \mathbb{E}[Y \otimes X]$ and $C_{XX} = \mathbb{E}[X \otimes X]$. In its concluding discussion it asks whether a nonlinear model $Y = f(X) + \varepsilon$ admits, under reasonable regularity assumptions, a finite β such that

$$C_{XY}C_{YX} \leq \beta C_{XX}^2. \tag{1}$$

This is equivalent to the relevant covariance range inclusion and to existence of a bounded linear least-squares minimiser.

——— Learning linear operators: Infinite-dimensional regression as an inverse problem ———

> **Existence of a solution, model misspecification, and covariance bounds.** Although we are able to give basic existence criteria of minimisers of the infinite-dimensional regression problem in terms of alternative characterisations of range inclusions of bounded linear operators based on ideas by Douglas (1966), the results formulated in Proposition 3.11, Remark 3.12 and Proposition 3.15 seem a bit unspecific in our context of covariance operators (we also refer to the recent discussion by Klebanov et al., 2021). In fact, the importance of range inclusion properties is already mentioned by Baker (1973) in his seminal paper on covariance operators. However, to the best of our knowledge, no analysis of functional models for X and Y satisfying this property are available in the literature. Under the assumption that $Y = f(X) + \varepsilon$ $\mathbb{P}$-a.s. for some nonlinear transformation $f\colon \mathcal{X} \to \mathcal{Y}$ and noise variable ε, does there exist some finite positive constant $\beta_{X,Y,f,\varepsilon}$ such that
>
> $$C_{XY}C_{YX} \leqslant \beta_{X,Y,f,\varepsilon} C_{XX}^2,$$
>
> where $\beta_{X,Y,f,\varepsilon}$ may depend on reasonable assumptions about smoothness and regularity properties of f and X, Y and ε and their joint distributions? Such a bound implies the range inclusion property due to Proposition 3.11(c) and hence describes classes of nonlinear infinite-dimensional statistical models which could still be reasonably empirically approximated by performing linear regression in a misspecified setting. For finite-dimensional X, this is always satisfied.

The following shows that even very strong smoothness, boundedness, and tail assumptions do not imply (1).

Theorem 1. *There are a compactly supported random variable X in ℓ_2 with injective covariance and a globally Lipschitz Fréchet-$\mathcal{C}^\infty$ map $f : \ell_2 \to \mathbb{R}$, all of whose derivatives are globally bounded, such that, for the bounded noiseless response $Y = f(X)$, no finite β satisfies (1). The same remains true after adding any independent, centred, square-integrable scalar noise to Y.*

Thus bounded (hence sub-Gaussian) data, moments of every order, smoothness of every order, and exogenous noise do not by themselves force the desired range inclusion.

2 A smooth map with nonsquare-summable sampled slopes

Let $(e_n)_{n\geq 2}$ be the standard basis of ℓ_2, and set

$$p_n = 2^{1-n}, \qquad a_n = \frac{1}{\log(n+1)}, \qquad r_n = \frac{1}{\sqrt{n}}, \qquad \rho_n = \frac{a_n}{4}.$$

Then $\sum_{n\geq 2} p_n = 1$, $a_n \downarrow 0$, and $(r_n) \notin \ell_2$. Choose $\eta \in \mathcal{C}_c^\infty(\mathbb{R})$ equal to one near zero and supported in $(-1,1)$, and define

$$f(x) = \sum_{n\geq 2} \sum_{\sigma\in\{-1,1\}} \sigma a_n r_n \, \eta\left(\frac{\|x - \sigma a_n e_n\|^2}{\rho_n^2} \right). \tag{2}$$

Lemma 1. *The supports in (2) are pairwise disjoint, and f is globally Lipschitz and Fréchet-$\mathcal{C}^\infty$ on ℓ_2. For every $k \geq 1$, $D^k f$ is globally bounded and $D^k f(0) = 0$.*

Proof. Opposite centres at index n are distance $2a_n$ apart. If $n \neq m$, the distance between any two centres is $(a_n^2 + a_m^2)^{1/2}$, whereas the sum of their support radii is

$$\frac{a_n + a_m}{4} \leq \frac{\sqrt{2}}{4}(a_n^2 + a_m^2)^{1/2}.$$

Thus the supports are disjoint. They are locally finite away from their only accumulation point, zero.

On the pair of supports indexed by n, the chain rule gives, for constants M_k depending only on k and η,

$$\|D^k f(x)\| \leq M_k a_n r_n \rho_n^{-k} \leq 4^k M_k r_n a_n^{1-k} = 4^k M_k \frac{(\log(n+1))^{k-1}}{\sqrt{n}} \longrightarrow 0.$$

Hence every derivative on $\ell_2 \setminus \{0\}$ extends continuously by zero at the origin. Induction using the Banach-space fundamental theorem of calculus makes these extensions the Fréchet derivatives of f at zero. For each fixed k the displayed sequence is bounded, so $D^k f$ is globally bounded. The case $k = 1$ implies global Lipschitz continuity. $\square$

In particular,

$$f(\sigma a_n e_n) = \sigma a_n r_n. \tag{3}$$

3 Covariances and the divergent test vectors

Define the symmetric atomic design by

$$\mathbb{P}(X = \sigma a_n e_n) = \frac{p_n}{2} \qquad (n \geq 2,\ \sigma \in \{-1,1\})$$

and put $Y = f(X)$. Both random variables are centred and bounded. Writing $\lambda_n = p_n a_n^2 > 0$, direct calculation from (3) gives

$$C_{XX} e_n = \lambda_n e_n, \qquad b := \mathbb{E}[YX] = \sum_{n\geq 2} \lambda_n r_n e_n. \tag{4}$$

For scalar Y, C_{YX} is represented by the vector b, and therefore $C_{XY} C_{YX} = b \otimes b$. Notice that C_{XX} is injective. The support closure $\{0\} \cup \{\pm a_n e_n : n \geq 2\}$ is compact because $a_n \to 0$.

Proof of Theorem 1. Suppose (1) holds. For $N \geq 2$, take the finite vector

$$x^{(N)} = \sum_{n=2}^{N} \frac{r_n}{\lambda_n} e_n.$$

Using (4),

$$\langle b, x^{(N)} \rangle = \sum_{n=2}^{N} r_n^2, \qquad \|C_{XX} x^{(N)}\|^2 = \sum_{n=2}^{N} r_n^2.$$

Evaluation of the positive-operator inequality on $x^{(N)}$ yields

$$\left(\sum_{n=2}^{N} r_n^2\right)^2 \le \beta \sum_{n=2}^{N} r_n^2, \quad \text{so} \quad \sum_{n=2}^{N} \frac{1}{n} \le \beta.$$

The harmonic partial sums diverge, a contradiction. Equivalently, the formal preimage of b under C_{XX} is (r_n), which is not in ℓ_2.

If independent centred noise ε is added, then $\mathbb{E}[\varepsilon X] = 0$, so C_{YX} and the same contradiction are unchanged. $\square$

4 Audit, novelty, and scope

The accompanying script checks probability normalisation, support separation, the derivative scales through order eight, and the exact finite-dimensional quadratic-form ratios. The proof itself is exact and does not rely on these computations.

A bounded search on 13 August 2026 used the exact title, the displayed inequality, and the phrases "covariance bounds," "range inclusion," and "nonlinear infinite-dimensional regression." It found the source and nearby operator-learning literature, but no later answer to this question or this construction. The novelty verdict is therefore *apparently new within the bounded search*, not an exhaustive bibliographic claim.

Because the source intentionally leaves "reasonable assumptions" open, the theorem does not exclude tailored sufficient hypotheses such as a finite-rank design or an explicit source/range condition. It gives a sharp negative result: arbitrary-order smoothness of f, bounded/sub-Gaussian inputs and outputs, injective covariance, and independent noise are still insufficient.

References

[1] M. Mollenhauer, N. Mücke, and T. J. Sullivan, *Learning linear operators: Infinite-dimensional regression as a well-behaved non-compact inverse problem*, arXiv:2211.08875v3 (2024), p. 31.